

ENGINEERING CALCULATION NOTE / REVISION B

# Pigjaw push-out fixture

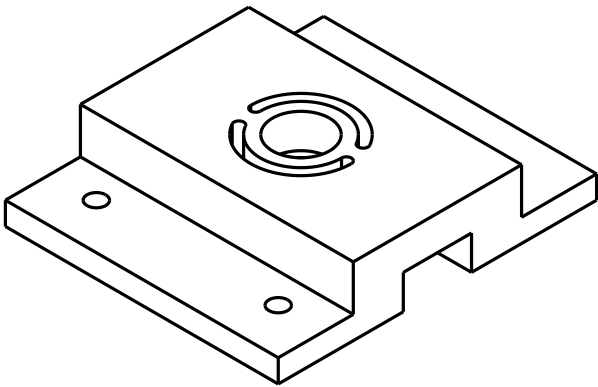
## Can printed PLA replace machined metal?

Hand calculations with retrospective PLA finite element results

Prepared for Jerich Lee

Units: N, mm, MPa

**Answer under the chosen screening assumptions.** This unchanged PLA geometry does **not** meet the selected 50  $\mu\text{m}$  maximum-annular-motion target at an assumed 250 N load. The final supplied FEA predicts **131.37  $\mu\text{m}$** , or **2.63 times the budget**. This establishes a modeled compliance concern; it does not establish fracture, a safe operating load, or that machined metal is mandatory.



Reference CAD supplied as Timmer\_NewPushoutforKershLab\_v2 v1.step.  
The figure and dimensional audit are retained from Revision A.

### Compare the responses at the same assumed force

| PLA result at 250 N                                | Downward displacement |
|----------------------------------------------------|-----------------------|
| Equivalent 60 mm bridge, hand calculation (page 5) | 31.77 $\mu\text{m}$   |
| Final M2 FEA, maximum on loaded annulus (page 10)  | 131.37 $\mu\text{m}$  |
| Analyst-selected maximum-annular-motion screen     | 50.00 $\mu\text{m}$   |

The bridge omits local ring and ligament deformation. The FEA resolves the slotted solid with idealized plate backing and washer hold-downs; it does not validate the bridge approximation. Aluminum and steel comparisons remain **hand-calculation scenarios only**.

### Keep the project history separate from the later analysis

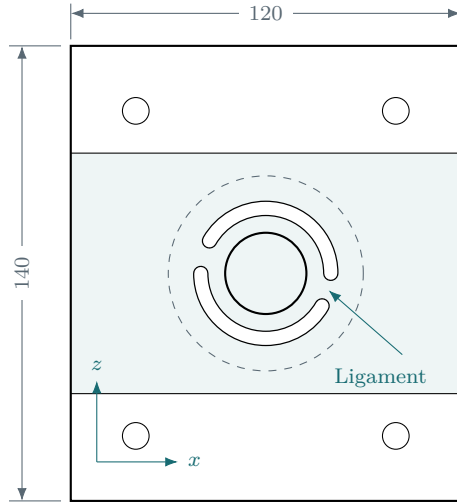
In the original 2025 project, the team did not use this concept because it feared PLA breakage. No observed fracture is established. The hand calculations and supplied FEA were added retrospectively on September 6–7, 2026; they did not drive the earlier decision.

**Document map:** geometry and assumptions, pp. 2–4; hand calculations, pp. 5–8; supplied FEA and numerical checks, pp. 9–13; final answer, p. 14; evidence and references, pp. 15–16.

## 01 / GEOMETRY

### Start with the actual load-bearing features

The STEP contains one solid. Its model coordinates have  $y$  through the thickness; the assumed test force acts downward along  $-y$ . The modeled solid is a reference for this calculation, not verified as-built geometry.[1]



| CAD-derived feature                    | Value                      |
|----------------------------------------|----------------------------|
| Overall $x \times z \times y$ envelope | $120 \times 140 \times 30$ |
| Raised deck extent in $z$              | 74                         |
| Mounting flange thickness              | 10                         |
| Central web, $h = 30 - 15$             | 15                         |
| Central opening, $D_h$                 | 25                         |
| Underside circular relief              | 60 dia.                    |
| Underside cross-channel width          | 30                         |
| Mounting holes                         | $4 \times 8.2$ dia.        |
| Mounting pattern in $x \times z$       | $80 \times 100$            |
| Slot inner / outer radii               | 17.85/22.15                |
| Slot width / end radius                | 4.30/2.15                  |
| Central annulus radial width           | 5.35                       |

Plan view, dimensioned in mm. Dashed circle denotes the 60 mm underside relief; slot linework is schematic.

#### The slot-end gap gives a local minimum width

Adjacent endcap centers in the  $x$ - $z$  plane are  $(17.3205, -10.0000)$  and  $(20.0000, 0.0265)$  mm. Their center distance minus the two end radii gives

$$b_\ell = \sqrt{(20.0000 - 17.3205)^2 + (0.0265 + 10.0000)^2} - 2(2.15) \simeq \boxed{6.0784 \text{ mm}}. \quad (1)$$

The calculation uses this minimum width as an equivalent rectangular ligament width. It is not a complete description of the curved load path.

**Assumed load path:** pusher  $\rightarrow$  implant  $\rightarrow$  interface / surrounding bone  $\rightarrow$  central annulus  $\rightarrow$  slot ligaments  $\rightarrow$  outer deck  $\rightarrow$  rigid base. Contact on the outer deck can bypass part of the annulus load; that possible load sharing is not quantified by this model.

**Do not use the 30 mm overall height as the bending thickness.** The central material above the underside relief is only 15 mm thick. For the same rectangular beam, substituting 30 mm would overpredict bending stiffness by  $(30/15)^3 = 8$ .

## 02 / LOAD AND FIT

## Define the force envelope and exit clearance

## What the supplied papers establish

Dewey et al. describe a critical-size porcine scaffold geometry of **25 mm diameter and 10 mm height**; their study is not a source of the present campaign's peak push-out force.[2] Lawson et al. validated a different, **80%-infill ABS** fixture only at loads  $\leq 250$  N, using smaller rat/rabbit calvarial specimens. That result is neither a PLA qualification nor a porcine force limit.[3]

Accordingly, this note uses  $F = 250$  N as a literature-context benchmark and  $F = 500$  and  $1000$  N as explicitly assumed sensitivity cases. The actual required force  $F_{\text{req}}$  remains an input to be established.

## A useful force conversion, not a strength prediction

For an idealized cylindrical bonded interface, its nominal lateral area is

$$A_i = \pi D_i t_i, \quad \tau_{\text{app}} = \frac{F}{\pi D_i t_i}, \quad F = \tau_{\text{app}} \pi D_i t_i. \quad (2)$$

Here  $t_i$  is the *effective axial contact length*, not fixture thickness  $h$ , and not automatically the scaffold's full height. This nominal stress follows the conventional push-out definition; local interface stress need not be uniform.[4]

| Assumed $F$ | $\tau_{\text{app}}$ for $D_i = 25$ , $t_i = 10$ mm | $\tau_{\text{app}}$ for $D_i = 25$ , $t_i = 3$ mm |
|-------------|----------------------------------------------------|---------------------------------------------------|
| 250 N       | 0.318 MPa                                          | 1.061 MPa                                         |
| 500 N       | 0.637 MPa                                          | 2.122 MPa                                         |
| 1000 N      | 1.273 MPa                                          | 4.244 MPa                                         |

The 3 mm and 10 mm contact lengths are sensitivity assumptions here. Do not import coated-metal implant strengths or Teflon press-fit retention forces as healed porcine scaffold strengths; Dewey 2019's Teflon-hole test addresses a different interface.[5]

## The 25 mm opening requires an independent clearance check

$$c = \frac{D_h - D_i}{2}. \quad (3)$$

For  $D_h = D_i = 25$  mm,  $c = 0$ . Dhert's model recommended a radial clearance of at least 0.7 mm; Lawson adopted that starting point for smaller specimens.[4, 3] Applying it provisionally gives

$$D_h \geq D_i + 2c = 25 + 2(0.7) = \boxed{26.4 \text{ mm}}, \quad (4)$$

before tolerances and alignment allowance. With a 25 mm hole, the corresponding implant diameter would be at most 23.6 mm.

**Conditional finding, not a universal design rule.** The actual implant diameter and release path must be measured. The 0.7 mm recommendation was not established for this 25 mm porous porcine construct. Enlarging the hole also narrows the annulus, so repeat the structural checks after any change.

## 03 / INPUTS

## Separate source properties from design assumptions

## Trial materials for the same geometry

| Material                      | $E$ (MPa) | $\nu$ | $\sigma_{\text{allow}}$ (MPa) | $\tau_{\text{allow}}$ (MPa) |
|-------------------------------|-----------|-------|-------------------------------|-----------------------------|
| Printed PLA surrogate         | 2000      | 0.35  | 10                            | 5                           |
| 6061-T6/T651 aluminum         | 68300     | 0.33  | 120                           | 60                          |
| 304 stainless, EN plate basis | 200000    | 0.30  | 105                           | 52.5                        |

**PLA:** UltiMaker’s printed, 100%-infill coupons have flexural moduli of approximately 2.74–3.02 GPa and upright tensile break strength  $33.1 \pm 2.8$  MPa, with no tensile yield reported for that orientation.[6] Here  $E = 2.0$  GPa and the 10/5 MPa normal/shear allowables are **analyst-selected screening assumptions**, not measurements of your print and not statistical lower bounds. The normal allowance corresponds to an assumed 20 MPa reference strength divided by two. The shear allowance is a separate conservative trial value, not a validated PLA failure law.

**Aluminum:**  $E = 68.3$  GPa follows Kaiser’s typical 6061 data.[7] A reference yield strength of 240 MPa is rounded down from the 35 ksi plate limit listed for the applicable thickness range; the trial normal allowance is  $240/2 = 120$  MPa.[8]

**Steel:** Outokumpu lists  $E = 200$  GPa and 210 MPa proof strength for Core 304 plate on its EN basis;  $210/2 = 105$  MPa is used here.[9] This is not a generic guarantee for every material sold as “304.” Confirm the purchasing specification and material certificate.

The metal shear allowances are half the normal allowances, a conservative pure-shear, Tresca-based screen. The Poisson ratios in the table are nominal modeling assumptions. For calculations,

$$G = \frac{E}{2(1 + \nu)}, \quad G_{\text{PLA}} = \frac{2000}{2(1.35)} = 740.7 \text{ MPa}. \quad (5)$$

For printed PLA this isotropic relationship is only a surrogate; interlayer shear stiffness may differ substantially.

## Explicit acceptance targets for this exercise

$$\boxed{\delta_{\text{fixture}} \leq 0.05 \text{ mm} = 50 \mu\text{m}}, \quad \sigma_{\text{screen}} \leq \sigma_{\text{allow}}, \quad \tau_{\text{screen}} \leq \tau_{\text{allow}}. \quad (6)$$

The  $50 \mu\text{m}$  displacement budget is chosen for this note, not imposed by the papers. Revision B applies it to the maximum downward movement on the FE loaded annulus. This screening metric is not automatically specimen-displacement error (page 10). The strength factor of two is already included in the trial allowables: do not apply it a second time to the same comparison.

All numerical checks assume short-duration, quasi-static loading and the adopted effective material properties. Unknown infill, print direction, defects, wet conditioning, temperature, creep, and load dwell prevent a final PLA qualification. The supplier data were obtained under stated room-temperature conditioning, not your complete laboratory exposure.[6]

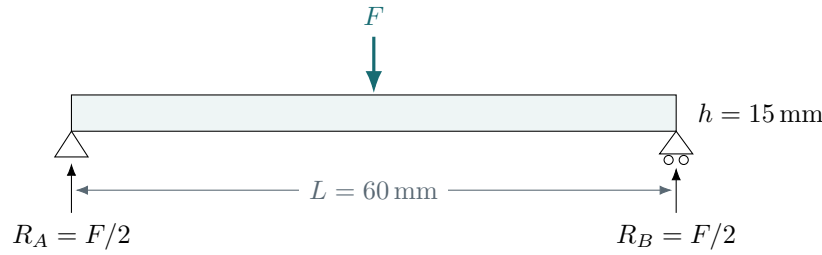
## 04 / GLOBAL MODEL

## Replace the deck by an equivalent bridge

Assume the bottom lands are backed by a rigid base, and take effective simple supports at  $z = \pm 30$  mm, beyond the circular relief. Use  $L = 60$  mm and  $h = 15$  mm. Remove the entire 44.3 mm-diameter slot envelope from the width, over the full span:

$$b_g = 120 - 2(22.15) = 75.7 \text{ mm}, \quad A_g = b_g h = 1135.5 \text{ mm}^2, \quad I_g = \frac{b_g h^3}{12} = 21290.6 \text{ mm}^4. \quad (7)$$

This is a **global bridge surrogate**, not a literal beam present in the CAD. It lumps load spreading into a central resultant and does not resolve the annulus or slot ligaments. Removing material reduces this surrogate's stiffness, but does not make it a rigorous upper bound on the real fixture's deflection.



## 1. Reactions, bending moment, and nominal stress

$$\begin{aligned} R_A = R_B &= \frac{F}{2}, & M_{\max} &= \frac{FL}{4}, \\ \sigma_{g,\max} &= \frac{M_{\max}(h/2)}{I_g} = \frac{3FL}{2b_g h^2}. \end{aligned} \quad (8)$$

At  $F = 250$  N:  $R_A = R_B = 125$  N,  $M_{\max} = 3750$  N mm, and

$$\sigma_{g,\max} = \frac{3(250)(60)}{2(75.7)(15)^2} = \boxed{1.321 \text{ MPa}}.$$

## 2. Bending plus transverse-shear deflection

Integrating the beam curvature with simple-support conditions gives  $\delta_b = FL^3/(48EI_g)$ . The shear-energy term gives  $\delta_s = FL/(4\kappa GA_g)$ , with the rectangular-section surrogate  $\kappa = 5/6$ :

$$\delta_g = \frac{FL^3}{48EI_g} + \frac{FL}{4\kappa GA_g}. \quad (9)$$

For PLA at 250 N,

$$\begin{aligned} \delta_b &= \frac{250(60)^3}{48(2000)(21290.6)} = 0.02642 \text{ mm}, \\ \delta_s &= \frac{250(60)}{4(5/6)(740.7)(1135.5)} = 0.00535 \text{ mm}, \\ \delta_g &= \boxed{0.03177 \text{ mm} = 31.77 \mu\text{m}}. \end{aligned} \quad (10)$$

Shear contributes about 17% of this total; omitting it is not ideal for the short  $L/h = 4$  bridge. Passing this check alone does not pass the slotted fixture.

## 05 / LOCAL CHECKS

### Check the annulus and its two narrow connections

Assume, for screening, that the entire test reaction enters the central annulus and then divides equally between its two ligaments. Actual specimen contact can instead share load with the outer deck; eccentric contact can also overload one ligament. The load partition must be checked physically.

#### 1. Average contact pressure: a necessary but weak check

Taking the annulus from  $r_i = 12.5$  to  $r_o = 17.85$  mm gives

$$A_c = \pi(r_o^2 - r_i^2) = \pi(17.85^2 - 12.5^2) = 510.1 \text{ mm}^2,$$

$$p_{\text{avg}} = \frac{F}{A_c} = \frac{250}{510.1} = \boxed{0.490 \text{ MPa}}. \quad (11)$$

An optimistic uniform-column shortening estimate is

$$\delta_c = \frac{Fh}{EA_c} = \frac{250(15)}{2000(510.1)} = 0.00368 \text{ mm}. \quad (12)$$

This excludes bending and assumes full-area contact. If only a fraction  $\eta$  contacts, replace  $A_c$  by  $\eta A_c$ . Neither  $p_{\text{avg}}$  nor  $\delta_c$  establishes specimen safety or fixture adequacy.

#### 2. Ligament shear: much less area carries the reaction

With  $b_\ell = 6.0784$  mm,  $h = 15$  mm, and  $V = F/2$  per ligament,

$$\tau_{\ell, \text{avg}} = \frac{F}{2b_\ell h} = \frac{250}{2(6.0784)(15)} = 1.371 \text{ MPa}. \quad (13)$$

For a rectangular-section peak factor of 1.5 and an *assumed* local concentration multiplier  $K_s = 2$ ,

$$\tau_{\ell, \text{screen}} = K_s \frac{3F}{4b_\ell h} = \boxed{4.113 \text{ MPa at } 250 \text{ N}}. \quad (14)$$

The ratio to the PLA trial allowance is  $4.113/5 = 0.823$ . A shear-only crossover occurs at  $F \approx 304$  N, but this is **not an assembly load rating**.  $K_s$  is a sensitivity choice, not a tabulated factor for this CAD.

#### 3. Local bending needs an effective moment arm

Represent each connection by a rectangular cantilever with resultant arm  $\ell$ . Then  $M_\ell = (F/2)\ell$  and, for an *assumed*  $K_t = 2$ ,

$$\sigma_{\ell, \text{screen}} = K_t \frac{3F\ell}{b_\ell h^2}. \quad (15)$$

| Assumed $\ell$ | $\sigma_{\ell, \text{screen}}$ at 250 N | Against PLA's 10 MPa allowance |
|----------------|-----------------------------------------|--------------------------------|
| 5 mm           | 5.48 MPa                                | Below allowance                |
| 10 mm          | 10.97 MPa                               | Slightly above allowance       |
| 15 mm          | 16.45 MPa                               | Above allowance                |

$\ell$  is **not a measured slot length**. It represents the unknown contact/resultant and restraint arrangement. The annulus can bend and twist, and the ligament roots are not ideal cantilever clamps. This sensitivity prevents a falsely precise PLA “pass.”

## 06 / COMPARISON

## The material gain is clear; the absolute rating is not

## Global bridge results for the declared 60 mm support span

| $F$ (N) | $\sigma_{g,\max}$ (MPa) | $\delta_g$ in $\mu\text{m}$ |          |           |
|---------|-------------------------|-----------------------------|----------|-----------|
|         |                         | PLA                         | Aluminum | Stainless |
| 100     | 0.528                   | 12.71                       | 0.371    | 0.126     |
| 250     | 1.321                   | 31.77                       | 0.928    | 0.316     |
| 500     | 2.642                   | 63.54                       | 1.856    | 0.631     |
| 1000    | 5.284                   | 127.08                      | 3.712    | 1.263     |

These are linear-elastic *global-model* estimates. They omit local ring motion and joints; higher-load PLA entries are extrapolations, not predictions after local damage. The nominal stress is the same for all materials in this statically determinate model. The allowables and deflections differ.

$$\frac{(EI)_{\text{Al}}}{(EI)_{\text{PLA}}} = \frac{68300}{2000} = \boxed{34.15}, \quad \frac{(EI)_{\text{steel}}}{(EI)_{\text{PLA}}} = \frac{200000}{2000} = \boxed{100}. \quad (16)$$

Steel gives about  $2.93\times$  aluminum's flexural stiffness at the same geometry. A large stiffness gain does not automatically justify steel if an aluminum design is later verified to satisfy the required accuracy with margin.

## Geometry and restraint can change the decision

| PLA sensitivity, all at 250 N          | Global $\delta_g$    | Meaning                      |
|----------------------------------------|----------------------|------------------------------|
| Baseline: $E = 2$ GPa, $L = 60$ mm     | $31.77 \mu\text{m}$  | Uses 64% of $50 \mu\text{m}$ |
| Stiffer print: $E = 3$ GPa, same $\nu$ | $21.18 \mu\text{m}$  | Still excludes local motion  |
| More flexible support: $L = 100$ mm    | $131.23 \mu\text{m}$ | Exceeds chosen budget        |

The 100 mm span is an alternate boundary-condition scenario. Although it equals the CAD mounting-row spacing, **bolt spacing does not establish a bearing span**. Actual support contact, not the nominal bolt pattern, determines the appropriate model.

For the baseline PLA bridge, compliance is

$$C_g = \frac{\delta_g}{F} = 1.2708 \times 10^{-4} \text{ mm/N}, \quad F_{\delta,g} = \frac{0.05}{C_g} = \boxed{393 \text{ N}}. \quad (17)$$

This 393 N value is only the point where the *global surrogate* uses the entire displacement budget. Local bending, shear, creep, and joints can set a lower usable force.

**A useful design lever:**  $\delta_b \propto L^3/(Ebh^3)$  and  $\delta_s \propto L/(Gbh)$ . Shortening the unsupported span or thickening the web can make a printed fixture more viable. Removing or repositioning slots can help only after confirming their assembly/alignment function.

## 07 / TEST QUALITY

## Survival is not the same as measurement fidelity

## Fixture compliance biases crosshead-based stiffness

For an ideal series load path, the same force passes through the specimen and fixture, but displacement is shared:

$$u_{\text{crosshead}} = u_s + FC_{\text{fixture}} + FC_{\text{machine}} + u_{\text{seating}}. \quad (18)$$

In a settled, linear interval, with total parasitic compliance  $C_p$ ,

$$\frac{1}{k_{\text{measured}}} = \frac{1}{k_s} + C_p, \quad \frac{k_s - k_{\text{measured}}}{k_s} = \frac{k_s C_p}{1 + k_s C_p}. \quad (19)$$

A maximum fractional stiffness error  $\varepsilon$  requires

$$k_p = \frac{1}{C_p} \geq \frac{1 - \varepsilon}{\varepsilon} k_s; \quad \varepsilon = 5\% \implies \boxed{k_p \geq 19k_s}. \quad (20)$$

**Example only:** for an assumed  $k_s = 500 \text{ N/mm}$ , the baseline PLA global bridge alone gives a 5.97% apparent stiffness reduction; the aluminum bridge gives about 0.185%. Other compliance adds to this. Neither  $k_s$  nor these errors was measured for the porcine specimens.

Pure elastic compliance does not by itself make an ideal in-line load cell read the wrong static force. It does affect inferred displacement, stiffness, energy, and the specimen's actual displacement rate. Interference, lateral force, slip, and changing contact are separate errors. Lawson specifically emphasizes coaxial alignment and avoiding unintended contact during push-out.[3]

## Mounting preload can exceed the test force

The STEP has 8.2 mm mounting holes, not a complete fastener specification. For an *illustrative* M8 fastener only, use the rough torque model  $T = KF_p d_b$  with assumed  $K = 0.20$ :

$$T = 2 \text{ N m} = 2000 \text{ N mm}, \quad d_b = 8 \text{ mm} \implies F_p \approx \frac{2000}{0.20(8)} = 1250 \text{ N}. \quad (21)$$

For a hypothetical 24 mm OD / 8.2 mm ID washer,

$$A_w = \frac{\pi}{4}(24^2 - 8.2^2) = 399.6 \text{ mm}^2, \quad p_w \approx \frac{1250}{399.6} = \boxed{3.13 \text{ MPa}}. \quad (22)$$

This is a sensitivity calculation, **not a recommended tightening torque**. Sustained pressure can cause a printed joint to relax. The FEA instead uses **16 mm OD patches with prescribed  $Y = 0$  and no preload**; this 24 mm washer example is not an FE input.

Support the lower lands on a verified rigid base and measure the fixture's own motion. Broad washers or suitable metal sleeves can protect printed mounting regions, but cannot eliminate deck or ligament bending. Pusher, stud, clamp, fastener, and base capacity must be checked separately; they are not contained in this single-solid STEP.

08 / FEA MODEL

A geometry-resolved follow-up to the hand screen

The following results were supplied from ANSYS Mechanical 2025 R2. They are screenshot readouts and recorded setup, not an independent solver rerun. The single fixture body uses an **isotropic linear-elastic PLA surrogate**,  $E = 2000 \text{ MPa}$  and  $\nu = 0.35$ , matching the hand-calculation inputs.<sup>[11]</sup>

Load and modeled contact

Uniform pressure acts downward ( $-Y$ ) on the top annulus, ID 25 mm / OD 35.7 mm:

$$A_c = \frac{\pi}{4}(35.7^2 - 25^2) = 510.1083 \text{ mm}^2,$$
$$F_{\text{ann}} = pA_c = 0.4901(510.1083) = 250.004 \text{ N} \simeq 250 \text{ N}.$$

(23)

The pressure supplies the whole test load; there is no second 250 N force. This annular-only loading is a **defined scenario**, not verified specimen contact or a demonstrated worst case. Actual contact could share load with the outer deck or load the ring eccentrically.

| Boundary condition | Recorded representation                                                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Plate backing      | Compression Only Support on the two lowest flat bottom lands; recess, channel ceiling, and hole/slot walls unsupported. Sliding friction neglected.              |
| Contact control    | Manual normal stiffness factor 10; update each iteration. This is a numerical contact factor, not a plate modulus.                                               |
| Washer hold-down   | Four flange-top annular patches, assumed OD 16 mm / ID 8.2 mm, total area about 593 mm <sup>2</sup> . $Y = 0$ ; $X, Z$ free. No washer solid or bolt pretension. |
| Locate A           | One bottom outer corner: $X = 0, Z = 0, Y$ free.                                                                                                                 |
| Locate B           | Other endpoint of the same $X$ -directed edge: $Z = 0, X, Y$ free. The points share $Y, Z$ and differ in $X$ .                                                   |

The user’s clarification that bolts and washers held the flanges to a flat rigid surface superseded the initial simply-resting assumption. The restraints approximate vertical hold-down and in-plane location; they do not reproduce real preload, washer bending, bolt stretch, or friction.

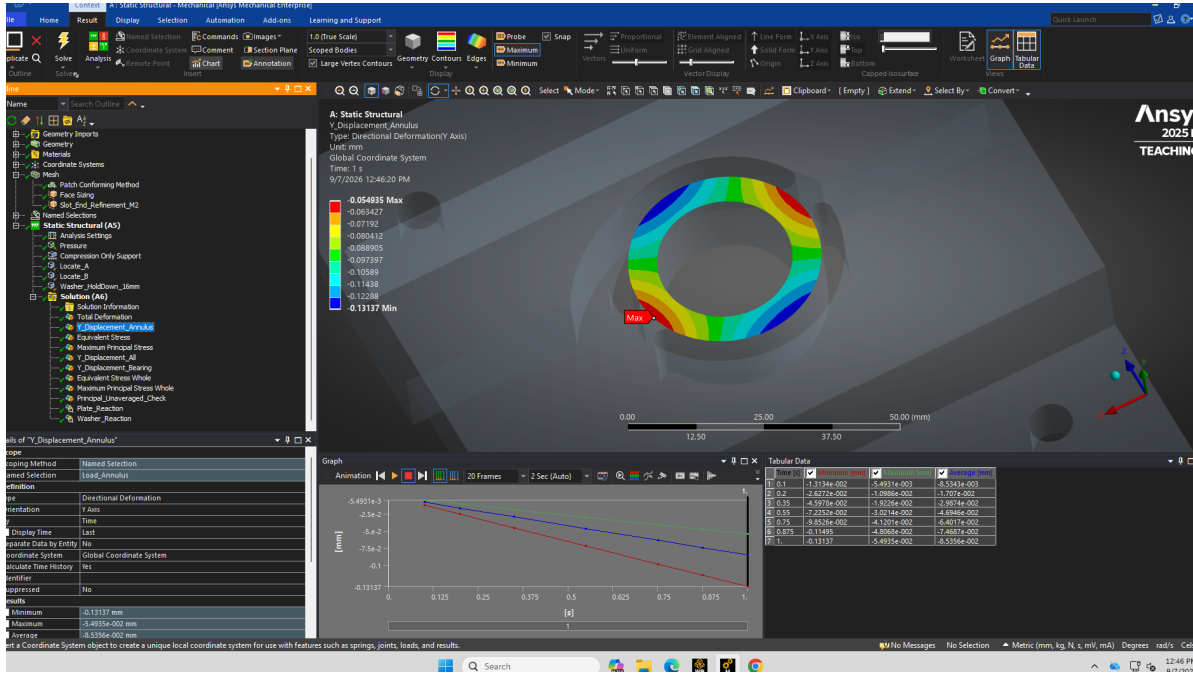
Analysis controls and reproducibility

The body is flexible; large deflection is off; no gravity is applied. The recorded setup uses one static load step, end time 1 s, automatic substeps (initial 10, minimum 5, maximum 100), program-controlled solver, and weak springs off. **Time = 1 s is the final static load-step parameter**, not a physical loading rate or dwell. Compression-only contact can open and close despite the linear material model.

The supplied STEP predates the annular and washer face partitions. No final ANSYS project, solver deck, result file, or full solver log was supplied. Some setup details are recorded workflow settings rather than independently audited inputs. Exact reproduction requires the final project or reconstruction of its partitions and settings.

## 09 / DISPLACEMENT

## Final PLA model exceeds the chosen motion budget



Supplied M2 screenshot, unmodified. Global Y directional deformation scoped to the loaded annulus; units mm; final static step. Displayed shape is visualization.

## Read the signed extrema correctly

The final annular minimum is  $u_{Y,\min} = -0.13137$  mm; the displayed maximum is  $u_{Y,\max} = -0.054935$  mm. Both are downward. The **most negative** value gives the largest downward movement:

$$\begin{aligned}\delta_{\text{ann,max}} &= |u_{Y,\min}| = 0.13137 \text{ mm} = 131.37 \mu\text{m}, \\ \frac{\delta_{\text{ann,max}}}{\delta_{\text{budget}}} &= \frac{0.13137}{0.05000} = 2.6274, \\ \delta_{\text{ann,max}} - \delta_{\text{budget}} &= 0.08137 \text{ mm} = 81.37 \mu\text{m}.\end{aligned}\quad (24)$$

The difference between annular extrema is  $0.076435$  mm =  $76.435 \mu\text{m}$ . It shows spatially nonuniform fixture movement; it does not by itself establish rigid-specimen tilt or interface deformation.

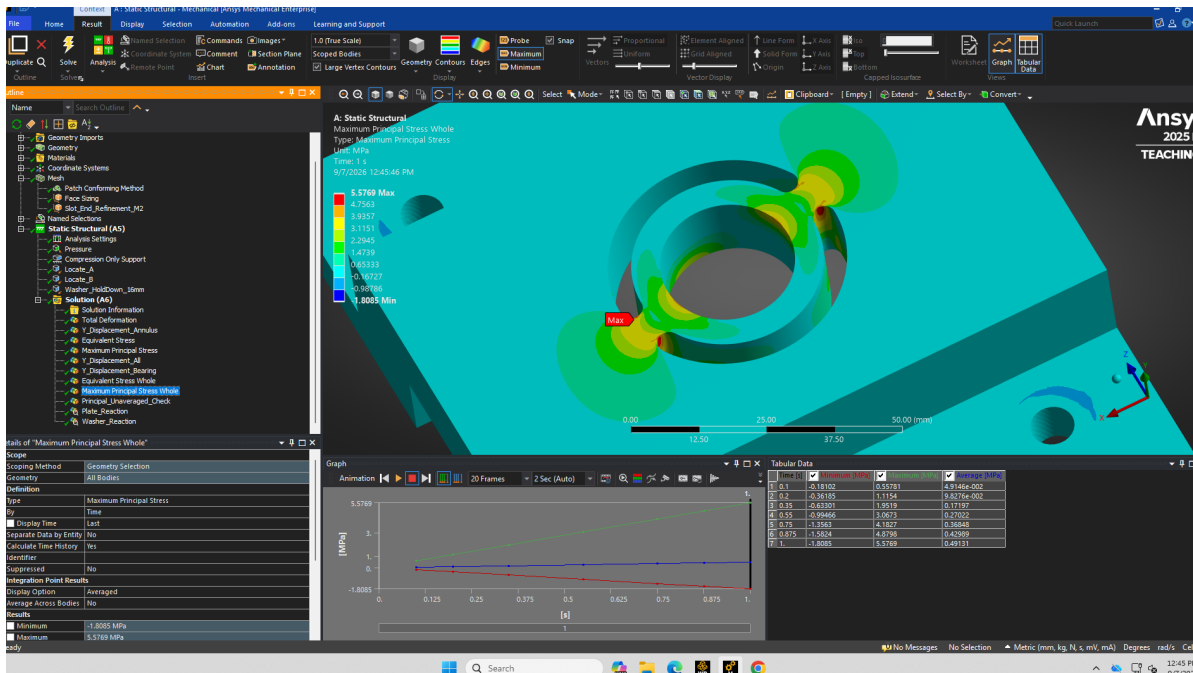
**Screening outcome: does not meet the chosen target.** At the assumed 250 N benchmark, this PLA model exceeds the  $50 \mu\text{m}$  maximum-annular-motion budget. The final local mesh refinement changed this maximum by only  $0.03 \mu\text{m}$  (page 12), which is small relative to the  $81.37 \mu\text{m}$  exceedance.

## What this displacement can and cannot mean

These are local fixture responses. Neither an extremum nor the mesh-dependent nodal arithmetic average is automatically specimen displacement, crosshead error, or a work-conjugate compliance. Do not use either as a scalar correction in the page 8 equations. Relating fixture motion to measurement accuracy requires the actual specimen contact and measurement arrangement, which are absent from this model.

## 10 / LOCAL STRESS

## Treat the tensile peaks as provisional indicators



Supplied M2 screenshot, unmodified. Maximum principal stress over the whole fixture, averaged display, units MPa; final static step.

| Maximum principal tensile stress | M1     | M2     |
|----------------------------------|--------|--------|
| Averaged peak (MPa)              | 5.4035 | 5.5769 |
| Unaveraged peak (MPa)            | 6.4151 | 6.0484 |

## Averaging changes the reported peak

Using the two final reported maxima,

$$\frac{\sigma_{1,\max}^{\text{unavg}} - \sigma_{1,\max}^{\text{avg}}}{\sigma_{1,\max}^{\text{avg}}} 100 = \frac{6.0484 - 5.5769}{5.5769} 100 = \boxed{8.45\%}. \quad (25)$$

The maxima can occur at different locations; this gap is not a local error estimate. Keep both values in the record rather than reporting only the smoother averaged peak.

**No complete strength pass or fracture conclusion.** Both tensile readouts are below the assumed 10 MPa normal allowance, but that comparison does not qualify a printed PLA part. Final von Mises, maximum shear, and minimum principal stress results were not supplied. Anisotropy, interlayer weakness, creep, damage, and cracking are absent from the model; the trial allowance is not a measured failure criterion.

The averaged peak changes by +3.21% from M1 to M2, while the unaveraged peak changes by −5.72%. The latter misses the chosen 5% successive-change criterion. In addition, the overlap of the lowest-quality element with the stress maximum is unresolved (page 12). **Local strength assessment therefore remains provisional**, even though displacement is stable in the final refinement.

## 11 / NUMERICAL CHECKS

### Stable displacement; incomplete stress convergence

All three levels use quadratic tetrahedra, Patch Conforming meshing, and adaptive sizing off. M0–M2 share the idealized bolted restraint and factor-10 contact control. The earlier unclamped run is a separate boundary-condition case.<sup>[11]</sup>

| Mesh | Sizing controls                                                                | Nodes     | Elements |
|------|--------------------------------------------------------------------------------|-----------|----------|
| M0   | 4 mm global; 1 mm annular face                                                 | 52,868    | 32,573   |
| M1   | 2 mm global; 1 mm annulus with 20 mm influence distance                        | 891,963   | 629,683  |
| M2   | Same as M1; four rounded slot-end walls at 0.5 mm with 4 mm influence distance | 1,321,643 | 941,817  |

M2 is a **local 0.5 mm slot-end refinement**, not a global 0.5 mm mesh.

| Reported response                       | M0           | M1       | M2       |
|-----------------------------------------|--------------|----------|----------|
| Maximum downward annular motion (mm)    | 0.12942      | 0.13134  | 0.13137  |
| Least downward annular motion (mm)      | 0.053713     | 0.054926 | 0.054935 |
| Averaged principal tensile peak (MPa)   | 4.6608       | 5.4035   | 5.5769   |
| Unaveraged principal tensile peak (MPa) | Not supplied | 6.4151   | 6.0484   |

### Successive changes, relative to M1

$$\begin{aligned}
 \Delta\delta\% &= 100 \frac{0.13137 - 0.13134}{0.13134} = \boxed{+0.02284\%}, \\
 \Delta\sigma_{1,\text{avg}}\% &= 100 \frac{5.5769 - 5.4035}{5.4035} = +3.209\%, \\
 \Delta\sigma_{1,\text{unavg}}\% &= 100 \frac{6.0484 - 6.4151}{6.4151} = -5.716\%.
 \end{aligned} \tag{26}$$

The absolute motion change is  $0.00003 \text{ mm} = 0.03 \mu\text{m}$ . This supports low sensitivity of the selected displacement metric to the final local refinement. It is not a formal error bound or proof of complete mesh independence.

### Contact and element-quality checks

On the coarse bolted mesh, switching from program-controlled contact to manual factor 10 changes maximum downward motion from 0.13024 to 0.12942 mm, a **0.630% decrease**. This is one contact-parameter sensitivity check; it does not establish all contact settings are insensitive.

M1 minimum/mean Element Quality is 0.23645/0.84389, retained from a handoff transcription; its original quality screenshot is not included. M2 is **0.077138/0.84497**. One element in the lowest histogram bin was isolated near a curved boundary; other low-quality bins exist. Coincidence with the stress peak was not established. This limits local stress confidence without, by itself, invalidating the displacement screen.

**Report precisely:** final annular displacement changed by about 0.023%; averaged peak stress changed by 3.21%; unaveraged peak stress changed by 5.72%. Do not label all stresses “fully converged.” Percentages inherit the rounded screenshot precision.

12 / EQUILIBRIUM AND COMPARISON

Check signed reactions and compare like loads

The final vertical reactions balance the applied load

The M2 force-reaction screenshots give the following global *Y* components at the final static step:

| Quantity                                    | Signed <i>Y</i> force (N) |
|---------------------------------------------|---------------------------|
| Plate reaction, compression-only support    | +355.55                   |
| Washer reaction, combined over four patches | −105.50                   |
| Sum of support reactions                    | +250.05                   |
| Nominal applied annular load                | −250.00                   |

$$r_Y = 355.55 - 105.50 - 250.00 = +0.05 \text{ N}, \quad \frac{|r_Y|}{250} 100 = \boxed{0.020\%}.$$

(27)

Using  $pA = 250.004 \text{ N}$  instead gives a residual of approximately  $0.046 \text{ N}$  ( $0.018\%$ ). Both indicate close vertical balance relative to  $250 \text{ N}$ . Values derive from rounded screenshot readouts and do not establish solver tolerances. This is a **vertical force-balance check**, not a full moment-balance or solver-log audit.

The downward washer reaction is compatible with the prescribed hold-down preventing local lift. It is the combined reaction of four constrained patches; it is neither an imposed bolt preload nor a per-bolt force. Add signed components, not reaction magnitudes.

FEA and the bridge do not predict the same quantity

| At 250 N, PLA     | Hand-calculation bridge                                   | Final M2 FEA                                                             |
|-------------------|-----------------------------------------------------------|--------------------------------------------------------------------------|
| Geometry          | Equivalent 60 mm span, reduced width, rectangular section | Slotted three-dimensional fixture body                                   |
| Restraint         | Ideal simple supports                                     | Compression-only plate backing, idealized washer hold-downs and locators |
| Reported response | Global bridge deflection                                  | Maximum downward nodal motion on loaded annulus                          |
| Displacement      | 0.03177 mm                                                | 0.13137 mm                                                               |

The FE readout is approximately  $4.14\times$  the bridge estimate at the same force. This difference exposes the limitation of the global surrogate, which omits the local annulus and ligament response. It is not a calibrated error factor, and the separate local hand models should not simply be added to manufacture an assembly prediction.

The original hand table also gives **0.12708 mm at 1000 N**. Its numerical proximity to **0.13137 mm at 250 N** is not agreement: both force and model differ. No aluminum or steel FEA was performed, and the metal hand-calculation scenarios on page 7 are not simulated or measured performance.

### 13 / FINAL ANSWER

## This PLA concept is not qualified by the present screen

### Could printed PLA replace machined metal?

**For this geometry and the chosen 250 N / 50  $\mu\text{m}$  screen: no.** The final model predicts 131.37  $\mu\text{m}$  maximum downward annular movement, exceeding the target by 81.37  $\mu\text{m}$ . The original hand-only result was insufficient to qualify the slotted part; the geometry-resolved follow-up identifies compliance as the clearest concern under the declared assumptions.

**That is not a universal rejection of printed PLA.** The actual force envelope, accuracy requirement, print properties, and specimen contact remain unknown. A lower-demand or redesigned printed fixture could be a candidate, but the supplied analysis does not establish its usable load or demonstrate fracture of this one.

**Machined 6061 aluminum remains a plausible stiffness alternative.** The same-geometry hand model gives about 34 times PLA's flexural stiffness; 304 steel gives about 100 times. These analytical comparisons motivate candidate selection. They do not establish that metal is mandatory, that either material meets the actual test requirement, or that a replacement was manufactured or tested. No metal FE run is included.

### What remains necessary before experimental use

1. **Set the requirement:** actual specimen-force envelope, equipment limits, displacement measurement location, and allowable measurement bias.
2. **Verify fit and load transfer:** implant diameter and exit travel, opening clearance, web/ligament dimensions, specimen contact footprint, rigid-base support, and real mounting hardware. A nominal 25 mm implant in a 25 mm opening has zero radial clearance.
3. **Resolve material and local strength:** actual print orientation, walls/infill, conditioning and dwell, relevant properties, and the provisional slot-end stress/mesh limitations. The current elastic FEA supplies no complete failure assessment.
4. **Check the complete assembly physically:** measure fixture motion and recovery under a representative dummy-specimen load within separately established limits. Include base, mounting, pusher, alignment and time-dependent response.

### Portfolio wording that preserves the chronology

During the original 2025 project, we set aside this fixture concept because of concerns that a printed PLA load path might fail. I revisited the unused concept in September 2026 by analyzing the supplied CAD with hand calculations and ANSYS Mechanical FEA. Under an assumed 250 N annular load, rigid-plate backing and idealized washer hold-downs, the final PLA model predicted about 0.131 mm maximum downward annular movement, above a chosen 0.050 mm screening target. Local mesh refinement changed that displacement by about 0.023%. The analysis identified a compliance concern under the stated assumptions; it did not demonstrate fracture or validate the fixture for use.

**Historical boundary:** the team feared breakage; no observed breakage is established. Whether this exact concept was printed, and what replacement was ultimately built or tested, remain unverified. Do not attribute the later analysis to the original 2025 decision or claim original authorship of the supplied reference CAD.

14 / EVIDENCE AND REVISION

A traceable record, with limits stated explicitly

What changed from Revision A

Revision A (September 6, 2026) contains the geometry audit and simplified hand calculations. Revision B (September 7, 2026) preserves their equations and scenario values, integrates the supplied M0–M2 PLA results, and updates the cover and decision to reflect the displacement screen. It adds model assumptions, stress limits, mesh/contact sensitivities, signed reaction balance, and chronology. It also distinguishes the illustrative 24 mm washer/preload calculation from the FE 16 mm restrained patches.

Evidence included with the editable bundle

| Record                    | Location and purpose                                                                                                                                                             |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Original handoff          | <code>evidence/handoff/</code> : handoff, model setup, result summary, evidence index, manifest and original verification script, retained unchanged.                            |
| Original calculation note | <code>evidence/handoff/sources/</code> : Revision A PDF and original source ZIP, preserved for comparison.                                                                       |
| CAD and screenshots       | Original STEP and 15 screenshots in the handoff’s <code>sources/</code> folder. The two figures reproduced here are the supplied M2 images.                                      |
| Editable Revision B       | <code>source/</code> : LaTeX, original analytical script and scenario outputs, CAD figure, and copies of all supplied screenshots.                                               |
| Revision checks           | <code>source/verify_revision.py</code> recomputes comparison arithmetic and checks consistency with the preserved handoff; <code>revision_checks.json</code> records the output. |

Which source supports which claim?

The geometry audit and hand-calculation source support pages 2–8. The recorded setup supports page 9 with the input-audit limitation stated there. M2 annular and principal-stress screenshots support pages 10–11. M0/M1/M2 mesh and result screenshots support page 12; the final plate and washer reaction screenshots support page 13. The handoff records the original decision and the retrospective chronology.

All **27 manifest-listed handoff files passed SHA-256 verification**. All 12 original scenario rows were independently recomputed across 72 numerical output fields with no mismatch. New percentages and equilibrium arithmetic were also checked. These checks establish file integrity and arithmetic consistency, not solver correctness or physical validation.

**Coverage and exclusions.** No final solver archive/input/results or full solver log was supplied. Some earlier M0/M1 reaction, bearing and M1 quality values exist only as handoff transcriptions. The final balance uses the M2 reaction screenshots. Missing M2 bearing penetration and unreported final stress components remain unknown, not zero. No real force–displacement measurements, print-specific properties, biological failure model, aluminum simulation, or demonstrated replacement build are established by this record.

The original STEP contains the supplied reference solid, not the later FE face partitions or complete test assembly. Bibliographic statements on the next page retain Revision A’s documented source readings; this revision does not create new experimental evidence from literature.

## 15 / REFERENCES

## Literature and source provenance

## References

- [1] **User-supplied CAD.** *Timmer\_NewPushoutforKershLab\_v2 v1.step*. Revision A records a single B-rep solid imported using CadQuery/Open CASCADE in millimeters; bounds, face levels, radii, hole positions and slot-end centers were extracted. Revision B retains that dimensional audit and CAD-derived figure. This is supplied reference geometry, not an authorship claim.
- [2] Dewey, M. J., et al. (2022). *Repair of critical-size porcine craniofacial bone defects using a collagen–polycaprolactone composite biomaterial*. *Biofabrication* 14, 014102. doi:10.1088/1758-5090/ac30d5. Supplied PDF, pp. 3–4 (article pp. 2–3): specimen geometry and study methods.
- [3] Lawson, Z. T., et al. (2021). *Methodology for performing biomechanical push-out tests for evaluating the osseointegration of calvarial defect repair in small animal models*. *MethodsX* 8, 101541. doi:10.1016/j.mex.2021.101541. Supplied PDF, pp. 3–6 and 10: ABS/infill, clearance, alignment, and the  $\leq 250$  N validation limit.
- [4] Dhert, W. J. A., et al. (1992). *A finite element analysis of the push-out test: influence of test conditions*. *J. Biomed. Mater. Res.* 26, 119–130. doi:10.1002/jbm.820260111. Supplied PDF, pp. 4–6 and 11 (one cover page precedes the article): nominal shear definition, rigid-support assumption, and interpretation limits.
- [5] Dewey, M. J., et al. (2019). *Shape-fitting collagen-PLA composite promotes osteogenic differentiation of porcine adipose stem cells*. *J. Mech. Behav. Biomed. Mater.* 95, 21–33. doi:10.1016/j.jmbbm.2019.03.017. Supplied offline manuscript, Sec. 2.5 / Fig. 4. Distinct initial-fit experiment; no force transferred to the present design requirement.
- [6] UltiMaker. *PLA Technical Data Sheet*, pp. 2–3. PDF filename v5.00; internal footer v2.00, April 20, 2022. Printed-coupon properties and conditioning. The lower adopted PLA stiffness/allowables in this note are assumptions, not supplier guarantees.
- [7] Kaiser Aluminum. *Rod & Bar Alloy 6061*, KA-RBH-6061-8.10, p. 1. Typical elastic modulus used for the alloy comparison.
- [8] Kaiser Aluminum. *Flat Rolled Products*, Table 7.2, KA-SP-AA6-1.09, PDF p. 20. 6061-T651 plate yield limit, 35 ksi, rounded down here to 240 MPa. Purchasing specification and stock certification remain necessary.
- [9] Outokumpu. *Core range datasheet*, Tables 5 and 7, pp. 8 and 10. Core 304 plate proof strength on EN basis and elastic modulus. Web references inspected September 6, 2026.
- [10] Jerich Lee. *Pigjaw Scaffold–Bone Interface Mechanical Testing* (portfolio), accessed September 6, 2026. Public project framing; not independent verification of experimental outcomes.
- [11] **User-supplied analysis handoff** (September 7, 2026). *HANDOFF.md*, *MODEL\_SETUP.md*, *results\_summary.json*, *EVIDENCE\_INDEX.md*, and retained ANSYS screenshots in *evidence/handoff/*. Original project chronology, recorded model setup, M0–M2 results and final scope. Screenshot readouts are not raw solver verification.

Reference links and source page locations are retained from Revision A. Rebuild instructions and complete local evidence paths are in the companion README. The hand calculations and supplied FEA are retrospective numerical screens, not a load rating.